Portfolio Theory

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Portfolio Theory

- Given a menu of assets, what do we invest in and how much?
  1. Asset allocation: what assets (e.g. bonds vs. stocks)
  2. Security selection: what securities (e.g. Apple vs. Google)
- How do we define best? Sharpe ratio
- Choose investment weights w_i to maximize portfolio Sharpe (Reward-to-Risk) ratio:

$$\max \frac{E[R^P] - R^f}{\sigma_P}$$

Portfolio Theory

- R_{it} is the return on asset i at time t
- w_i weight in asset i
- Portfolio return at time t

$$R_t^P = w_1 \cdot R_{1t} + \cdots + w_n \cdot R_{nt}$$

Portfolio Statistics

Given historical data

- $E[R^P] = \text{average}(R_t^P)$
- $\sigma_P = \text{stdev.s}(R_t^P)$

- Analytical formulas for two assets
  - $E[R^P] = w_1 E[R_1] + w_2 E[R_2]$
  - $\sigma_P = \sqrt{\sigma_P^2}$
  - $\sigma_P^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \sigma_{12}$
  - $\sigma_{12} = \rho_{12} \sigma_1 \sigma_2$

Diversification & Correlation

- Choose investment weights w_i to maximize portfolio Sharpe (Reward-to-Risk) ratio:

$$\max \frac{E[R^P] - R^f}{\sigma_P}$$

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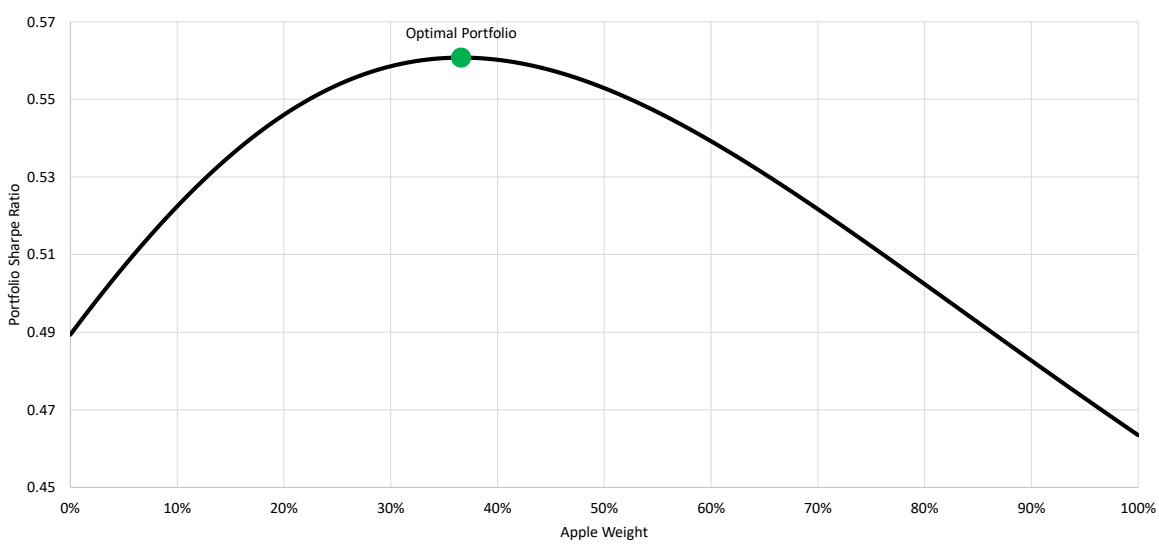
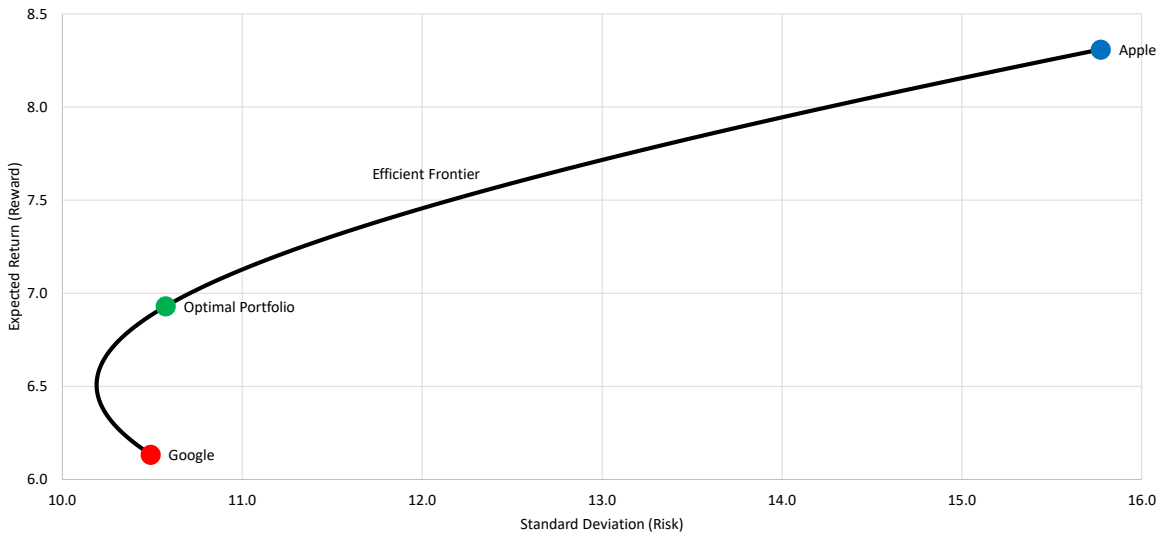
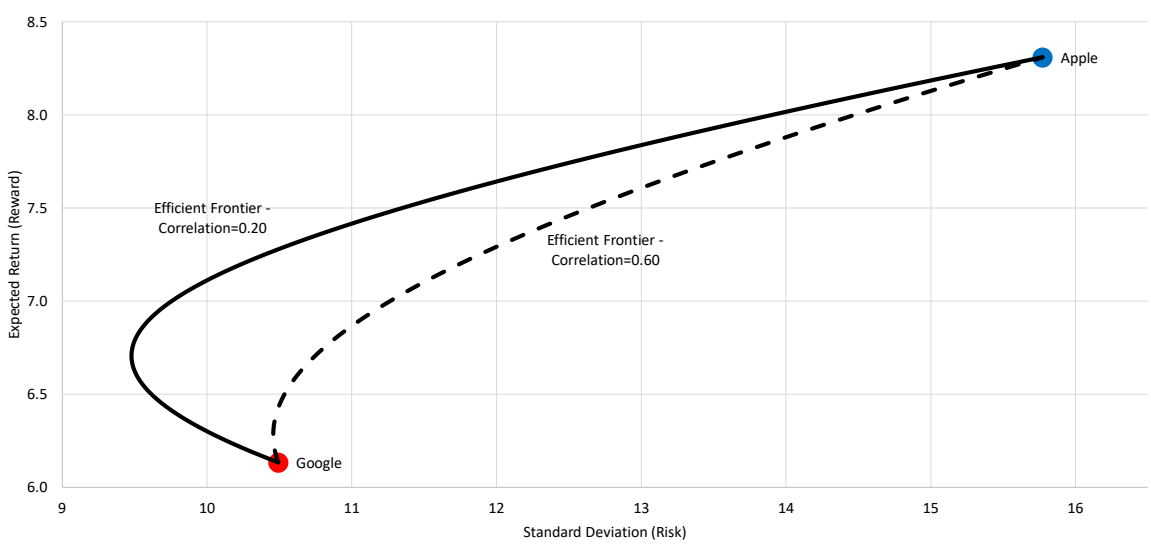
Security Selection



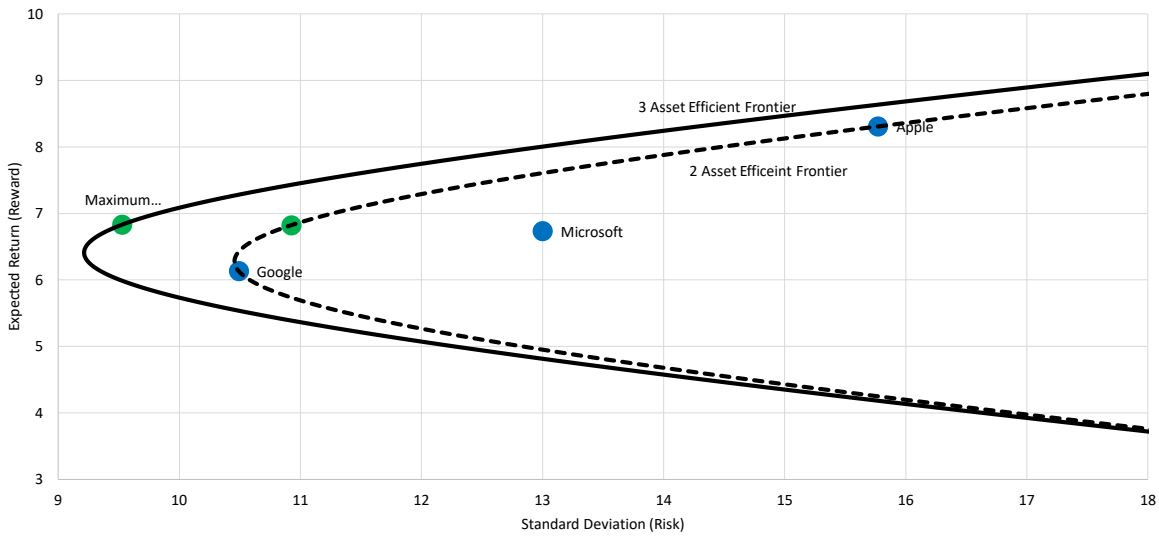
Security Selection



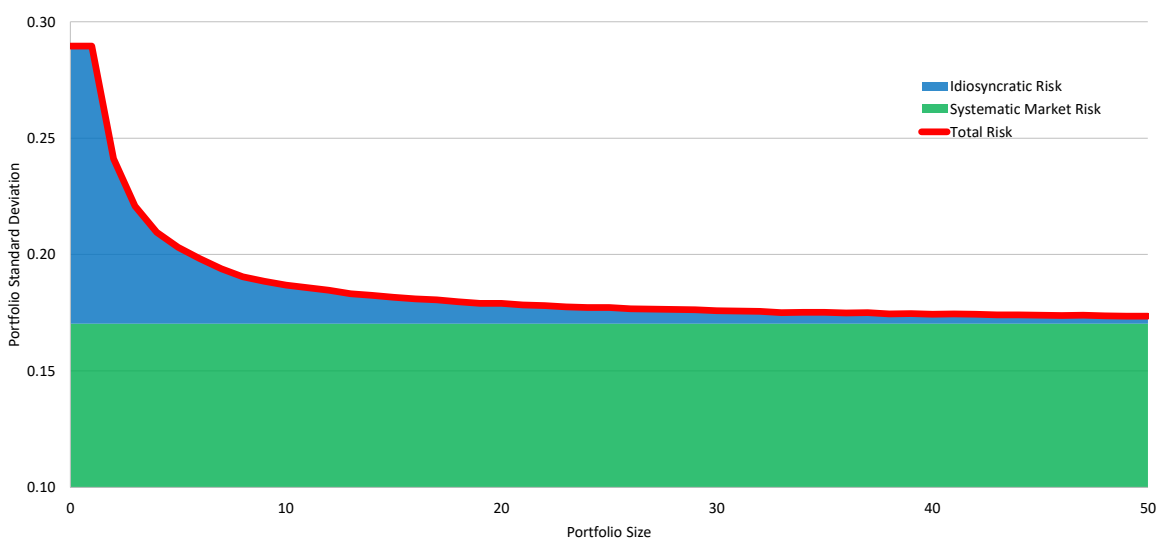
Diversification & Correlation



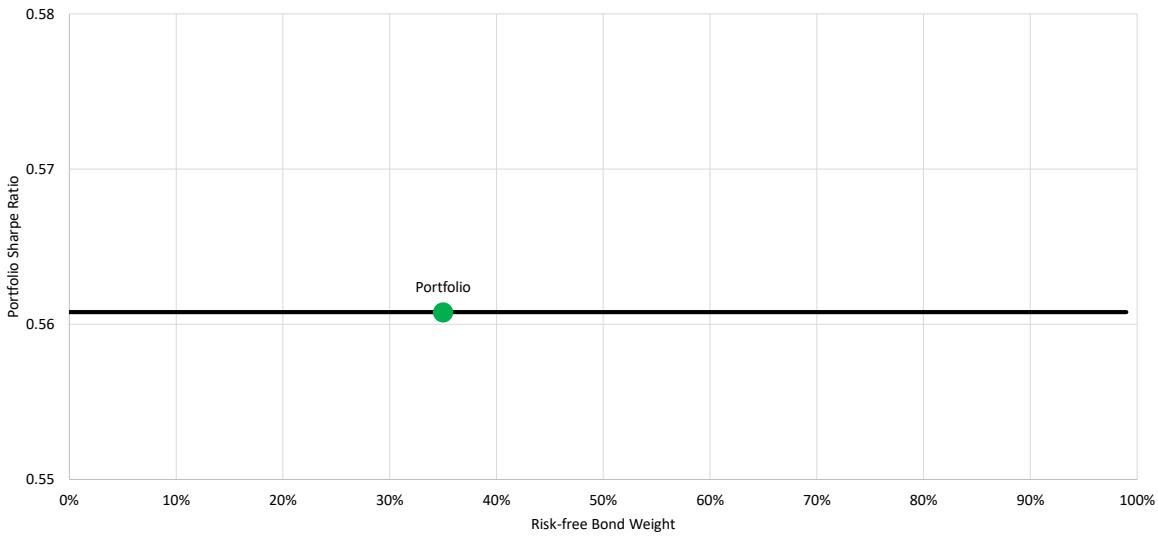
Diversification & Number of Assets



Diversification & Optimal Number of Assets



Capital Allocation



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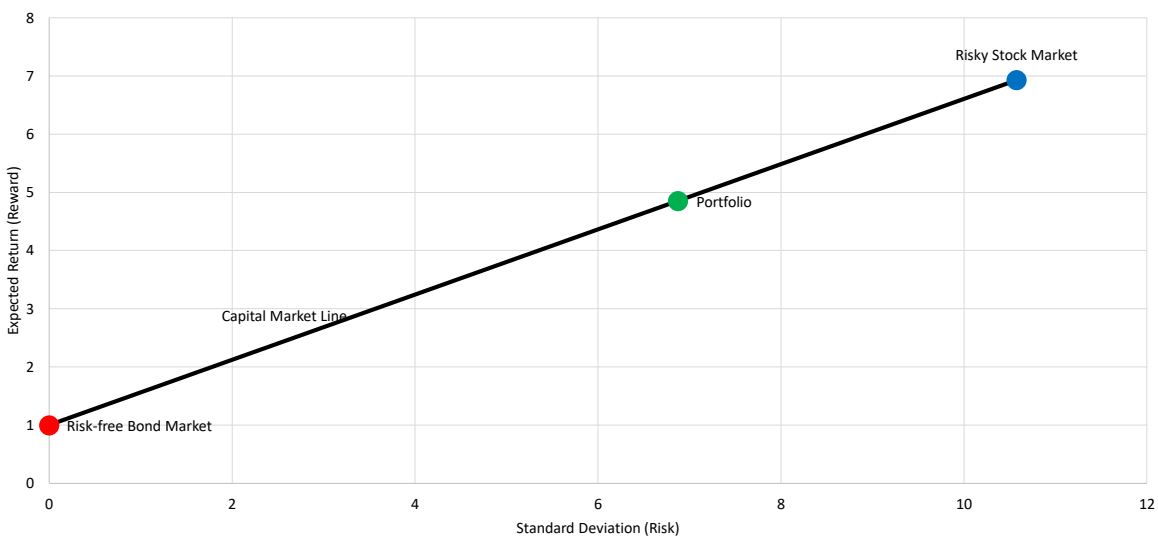
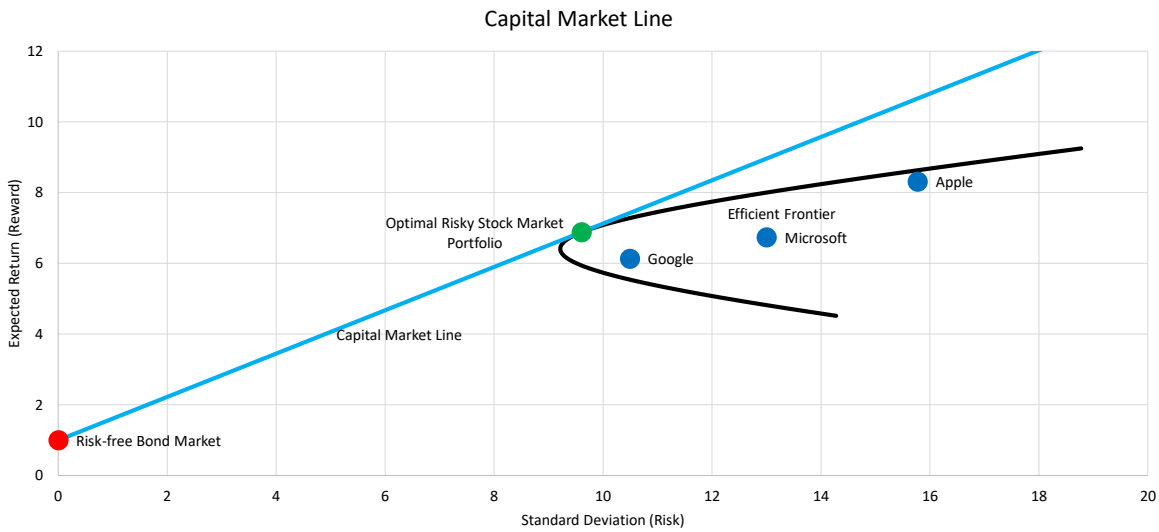


Portfolio Optimization



Portfolio Optimization

Summary

- Security selection
  - Calculate optimal risky market portfolio
  - Same for all investors
  - Only a function of assets' riskiness & correlation structure
- Asset allocation
  - Choose personal portfolio as a weighted average of risky market portfolio and risk-free bond market
  - Unique to each investors' risk preference

Portfolio Optimization

Summary

- Diversification allows us to obtain portfolios with higher Sharpe ratios relative to the portfolio's holdings.
- When assets are not perfectly correlated, the riskiness of the portfolio falls, resulting in a higher Sharpe ratio.
- Intuition: If two assets are negatively correlated, when one does poorly, the other does well on average. The assets hedge each other's risk, thus decreasing the overall risk of the portfolio.

Portfolio Theory & CAPM: History

- Economics:
  - Harry Markowitz 1952
  - William Sharpe 1964
  - Nobel Prize 1990 for Portfolio Theory & CAPM
- Industry:
  - Jack Bogle
  - Founded Vanguard in 1976 with the Vanguard 500 Index Fund
  - Benefits of diversification
  - $4.4 trillion AUM